

Contactless Fingerprint Quality Assessment System

1. Objective

The objective of this project is to develop a **Computer Vision-based quality assessment system** for contactless fingerprint images captured using a smartphone camera. Before a fingerprint image proceeds to biometric authentication, the system evaluates its quality to determine whether it is suitable for further processing.

The implemented quality assessment pipeline checks multiple image characteristics and provides an overall quality score along with user guidance for improving image capture.

2. Problem Statement

Unlike traditional contact-based fingerprint scanners, contactless fingerprint images captured using mobile cameras suffer from several challenges, including:

- Motion blur
- Poor lighting conditions
- Glare caused by reflections
- Partial fingertip capture
- Poor ridge visibility

These issues reduce the accuracy of downstream biometric processes such as feature extraction and fingerprint matching.

Therefore, a quality assessment stage is essential to reject low-quality images before authentication.

3. Methodology

The implemented system performs the following quality checks:

Blur Detection

Image sharpness is evaluated using the **Variance of Laplacian** method. Low variance indicates blurred images.

Brightness Analysis

The average grayscale intensity is computed to determine whether the image is too dark or too bright.

Glare Detection

Overexposed pixels are detected to estimate glare caused by excessive lighting or reflections.

ROI (Region of Interest) Completeness

Otsu thresholding and contour detection are used to estimate whether a sufficient portion of the fingertip is present in the captured image.

Ridge Clarity

A Gabor filter is applied to enhance directional ridge patterns. The variance of the filter response is used to estimate ridge clarity.

Composite Quality Score

All quality metrics are normalized and combined using weighted scoring to generate a composite quality score. Images that satisfy both the quality threshold and individual quality constraints are marked as **PASS**, while others receive **FAIL** along with guidance for improvement.

4. Technologies Used

- Python
 - OpenCV
 - NumPy
 - Pandas
 - Streamlit
 - Git
 - GitHub
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5. Results

The developed system successfully evaluates fingerprint image quality using multiple quality metrics and provides:

- PASS/FAIL quality decision
- Composite quality score
- Individual metric analysis
- User guidance
- Interactive Streamlit dashboard
- Batch dataset evaluation with CSV report generation

The modular architecture also makes the system easy to extend with additional biometric processing stages.

6. Future Improvements

Future enhancements may include:

- Automatic finger detection
- Liveness detection
- Deep learning-based segmentation
- CLAHE-based image enhancement

- Minutiae extraction
 - ISO/IEC 19794-4 template generation
 - Fingerprint matching
 - Mobile deployment
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7. Conclusion

This project demonstrates how computer vision techniques can be used to evaluate the quality of contactless fingerprint images before biometric authentication. The implemented quality assessment pipeline improves the reliability of downstream biometric processing by rejecting unsuitable images and providing actionable user feedback.

Report Answers

1. What threshold did you set for blur? How did you decide?

Answer

Blur Threshold: 10.0

The blur threshold was selected through empirical testing (trial and error) on my sample fingerprint images.

Initially, I tested a threshold of **5.0**, but it incorrectly accepted several slightly blurred images. Increasing the threshold to **10.0** produced more consistent results by correctly distinguishing between sharp and blurred fingerprint images while minimizing false positives.

Testing was performed using:

- Original high-quality fingertip images
- Artificially blurred versions created using Gaussian Blur in OpenCV
- Multiple comparisons of Laplacian variance scores

This value provided the best balance for my current dataset.

2. Which metric was hardest to implement correctly?

Answer

The most challenging metric was **ROI (Region of Interest) Completeness**.

Initially, simple thresholding produced noisy binary masks, causing incorrect contour detection and inaccurate ROI measurements.

To improve the results, I:

- Applied Gaussian Blur before thresholding.
- Used Otsu Thresholding to automatically determine the threshold value.
- Performed morphological opening to remove small noise.
- Selected only the largest contour as the fingertip region.

These improvements significantly increased the reliability of ROI estimation.

3. What is NFIQ2? Why is it not reliable for contactless images?

Answer

NFIQ2 (NIST Fingerprint Image Quality 2) is a standardized fingerprint quality assessment algorithm developed by NIST for evaluating images captured using **contact-based fingerprint scanners**.

It estimates whether a fingerprint image is likely to produce accurate matching results.

However, NFIQ2 is not directly suitable for smartphone-based contactless fingerprint images because:

- Contactless images suffer from perspective distortion.
- Lighting conditions vary significantly.
- Mobile cameras capture curved finger surfaces.
- Resolution and scale differ from 500 DPI scanner images.
- Background information is present.

Therefore, additional quality checks specifically designed for contactless fingerprint images are required before biometric processing.

4. Name three additional quality checks you would add.

Answer

If this system were deployed in production, I would include:

1. Finger Angle Detection

Reject images where the fingertip is excessively rotated or tilted.

2. Finger Distance Estimation

Ensure the finger is neither too close nor too far from the camera.

3. Wet/Dry Finger Detection

Detect moisture, excessive dryness, or dirt that can reduce ridge visibility and negatively affect feature extraction.

5. Agricultural workers may have worn fingerprints. What should the system do?

Answer

A fixed ridge clarity threshold may unfairly reject users whose fingerprints are naturally worn due to manual labor.

Instead, the system should adopt an adaptive quality assessment strategy.

Possible improvements include:

- Using personalized thresholds after successful enrollment.
- Combining multiple quality metrics instead of relying only on ridge clarity.
- Capturing multiple images and selecting the best one.
- Providing user guidance for repositioning the finger before rejection.
- Allowing human verification or an alternative biometric method if quality remains consistently low.

This approach improves accessibility while maintaining security.